

5.1 Mathematical Induction - 1

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10:19 AM

Mathematical induction is another method of writing proofs. It is particularly useful when proving a statement is true for all integers $n \in [n_0, \infty)$. Mathematical induction is an argument structured in the following way...

"I can climb an infinite staircase."	}	Claim
I can climb the first step.	}	Base Case
If I can reach some future step, there is a step after it that I can climb.	}	Inductive Hypothesis/Step
Therefore I can climb the infinite staircase.	}	Conclusion

Principle of Induction: To prove that $P(n)$ is true for all positive integers n ...

- 1) Verify $P(1)$ is true. Base Case
- 2) Assume $P(k)$ is true for some fixed positive integer k . Inductive Hypothesis
- 3) Prove the implication $P(k) \Rightarrow P(k+1)$ is true. Inductive Step.

Argument Form:

$$\left[P(1) \wedge (\forall k [P(k) \Rightarrow P(k+1)]) \right] \Rightarrow \forall n P(n)$$

Another way to view induction is as a line of dominoes. The base case tips the first one, setting off a chain reaction.

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first one, setting of a chain reaction.

Then if/when any domino in this line is knocked over, it will knock down the next one. This guarantees that eventually, all dominoes get knocked over.

The base case is not always $n=1$.

Always read the statement and decide what the first result would be.

* $\left\{ \begin{array}{l} \text{There is an excellent template for} \\ \text{proofs by induction on page 335.} \\ \text{Quizzes \& exams will expect this format.} \end{array} \right.$

Show that for any positive integer n

$$1+2+\dots+n = \frac{n(n+1)}{2}.$$

Let $P(n)$: " $1+2+\dots+n = \frac{n(n+1)}{2}$ " where

n is any positive integer. Then the smallest value of n in consideration is 1.

Base Case ($n=1$): $P(1)$: " $1 = \frac{1(1+1)}{2}$ ".

$\frac{1(1+1)}{2} = \frac{2}{2} = 1$. Thus $P(1)$ is true.

Inductive Step: Assume $P(k)$ is true

for some arbitrary integer $k \geq 1$. Then

$$1+2+\dots+k = \frac{k(k+1)}{2}.$$

We want to show that then $P(k+1)$ is also true, that is

$$1+2+\dots+(k+1) = \frac{(k+1)(k+2)}{2}.$$

Notice

$$1+2+\dots+(k+1) = 1+2+\dots+k + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1) \quad \text{by } P(k).$$

$$\text{Then } \frac{k(k+1)}{2} + (k+1) = \frac{k(k+1)}{2} + \frac{2(k+1)}{2}$$

$$= \frac{k^2+k+2k+2}{2} = \frac{k^2+3k+2}{2}$$

$$= \frac{(k+1)(k+2)}{2}.$$

So $P(k+1)$ is true. This completes the inductive step.

Therefore, by mathematical induction

$$1+2+\dots+n = \frac{n(n+1)}{2} \text{ for all positive}$$

integers n

Prove that for any positive integer n
 $1+3+5+\dots+(2n-1) = n^2$.

Let $P(n)$: " $1+3+\dots+(2n-1) = n^2$ " for any positive integer n . Then the smallest value of n in consideration is 1.

Base Case ($n=1$): $P(1)$: " $1 = (1)^2$ " so

$P(1)$ is true.

Inductive Step: Assume $P(k)$ is true for some arbitrary positive integer $k \geq 1$. Then

$$1+3+\dots+(2k-1) = k^2.$$

We want to show that then $P(k+1)$ is also true, that is

$$1+3+\dots+[2(k+1)-1] = (k+1)^2.$$

Notice

$$\begin{aligned} 1+3+\dots+[2(k+1)-1] &= 1+3+\dots+(2k-1) + [2(k+1)-1] \\ &= k^2 + [2(k+1)-1] \text{ by } P(k). \end{aligned}$$

Then

$$k^2 + [2(k+1)-1] = k^2 + 2k + 1 = (k+1)^2$$

so $P(k+1)$ is true. This completes the inductive step.

Therefore by mathematical induction,

$$1+3+\dots+(2n-1) = n^2 \text{ for any positive}$$

integer n

Prove that for any integer greater than 6,
 $3^n < n!$

Let $P(n)$: " $3^n < n!$ " where n is an integer

$$= (k^5 - k) + 5k^4 + 10k^3 + 10k^2 + 5k$$

where $k^5 - k$ is divisible by 5 by $P(k)$

and $5k^4 + 10k^3 + 10k^2 + 5k$ is divisible by 5.

Thus all of $(k^5 - k) + 5k^4 + 10k^3 + 10k^2 + 5k = (k+1)^5 - (k+1)$ is divisible by 5. So $P(k+1)$ is true. This completes the inductive step.

Therefore by mathematical induction, 5 divides $n^5 - n$ for any non-negative integer n . $\square$